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Class: T12 **Roll No.:** 39

EXPERIMENT 8

Title: Implementation of an AWS CI/CD Pipeline using AWS CodePipeline, CodeBuild and Amazon S3

Aim

To design and implement a continuous integration and continuous deployment (CI/CD) pipeline using AWS CodePipeline, AWS CodeBuild, GitHub, and Amazon S3, so that updates pushed to the source repository are automatically built and deployed to an S3-hosted static website.

Theory

Continuous Integration (CI) is the practice of frequently integrating code changes into a shared repository and validating them through an automated build process. **Continuous Deployment (CD)** automates the release of validated changes to the target environment. Together, CI/CD reduces manual effort, improves consistency, and helps deliver updates faster.

AWS CodePipeline is a managed service that coordinates the stages of a software release workflow. It can connect a source provider such as GitHub to a build service and a deployment target. **AWS CodeBuild** compiles, tests, or packages the application based on a build specification file such as *buildspec.yml*. **Amazon S3** can host static website files and store the artifacts or deployed output. GitHub acts as the source repository, and a source change can trigger the pipeline automatically.

Workflow: Developer pushes code to GitHub → CodePipeline detects the change → CodeBuild runs the build instructions → output is deployed to the configured S3 bucket → the updated static website is available.

Procedure / Implementation

1. Create an Amazon S3 bucket for the website artifacts and configure the required website settings.
2. Create a CodeBuild project and connect it to the GitHub repository.
3. Configure the build specification file (buildspec.yml) for the project.
4. Create a CodePipeline pipeline with GitHub as the source stage, AWS CodeBuild as the build stage, and Amazon S3 as the deploy stage.
5. Configure the pipeline's service role, artifact store, repository, and branch.
6. Run the pipeline and verify that the Source, Build, and Deploy stages complete successfully.
7. Update a website file in GitHub and confirm that the pipeline runs again and the deployed website reflects the update.

Relevant Aim

To demonstrate automated deployment of a static website by integrating GitHub, AWS CodePipeline, AWS CodeBuild, and Amazon S3, and to verify that source-code updates are automatically reflected on the hosted website.

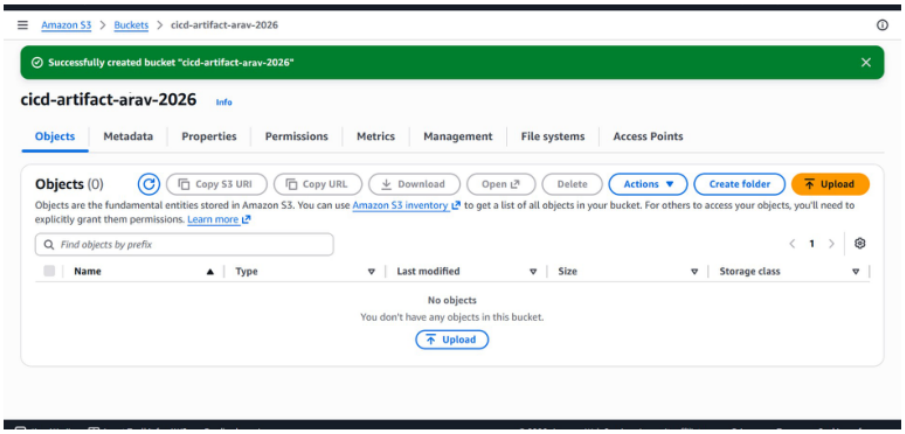
Result

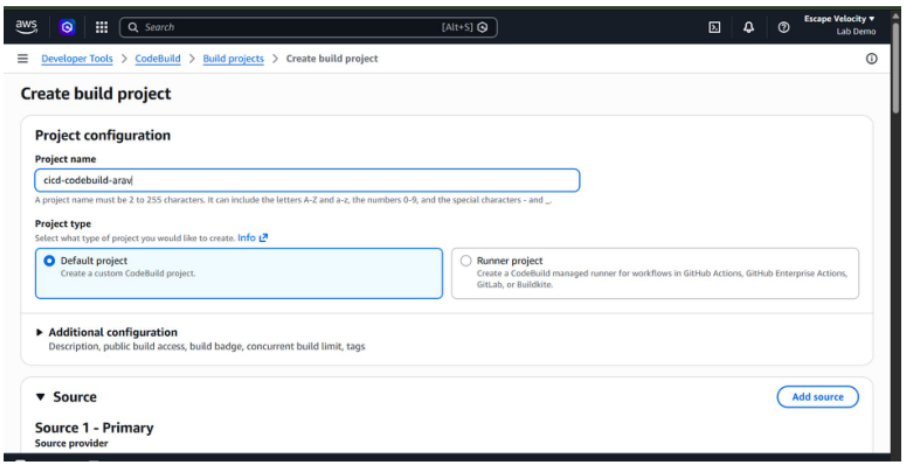
The AWS CI/CD pipeline was configured and executed successfully. The Source, Build, and Deploy stages completed, and an update to the repository was reflected on the S3-hosted website, demonstrating automated deployment.

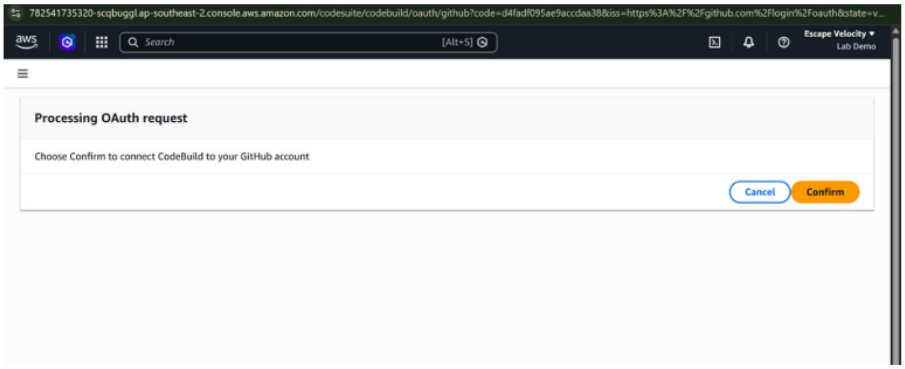
Screenshots / Evidence

The following screenshots are retained from the supplied experiment record.

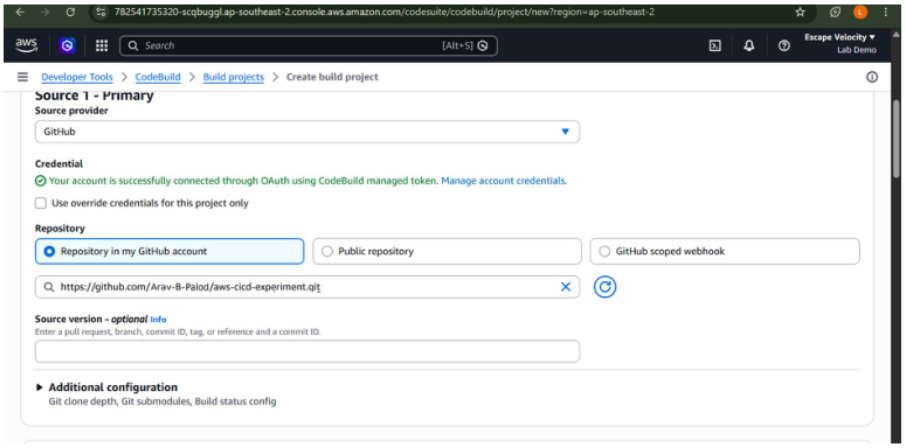
Experiment Evidence — Original Page 1

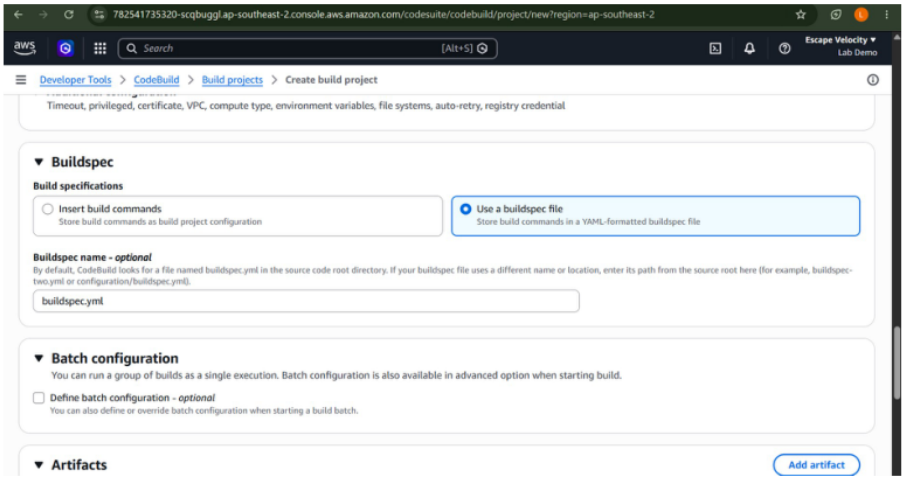


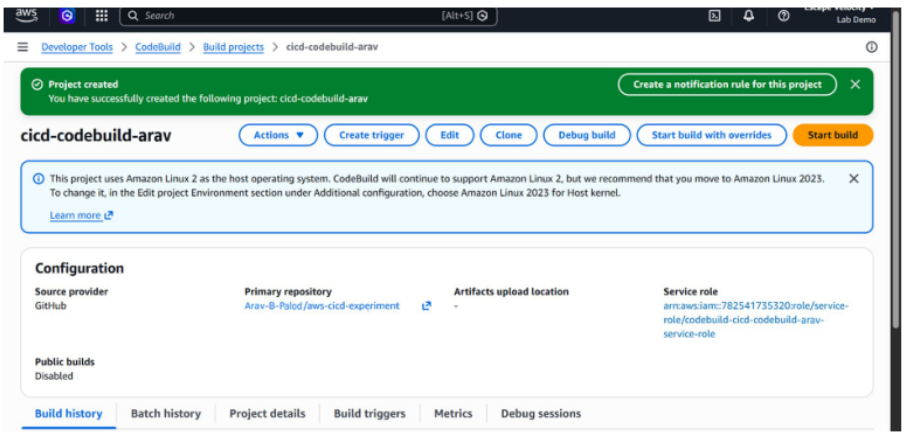




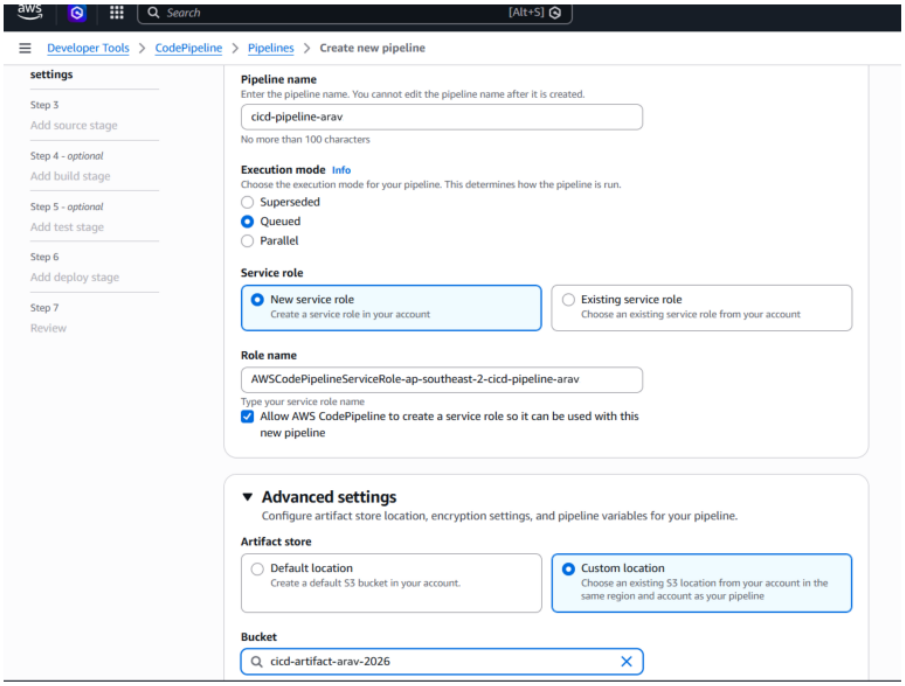
Experiment Evidence — Original Page 2

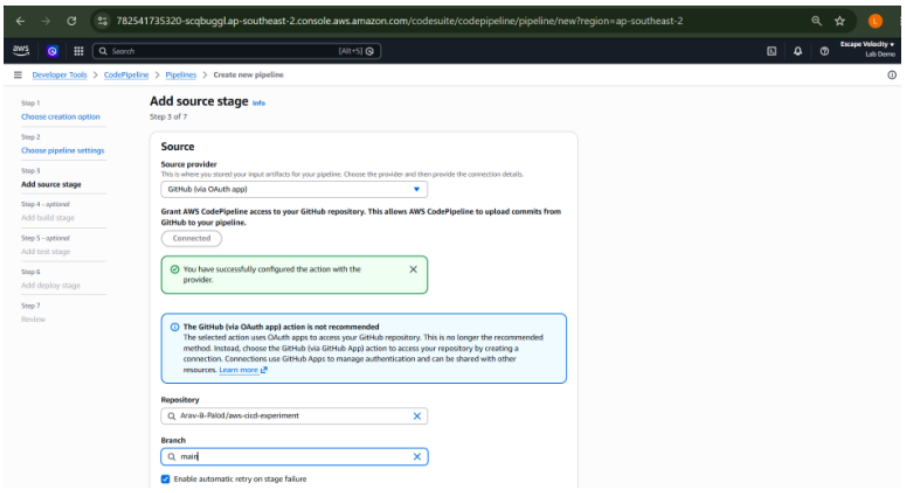




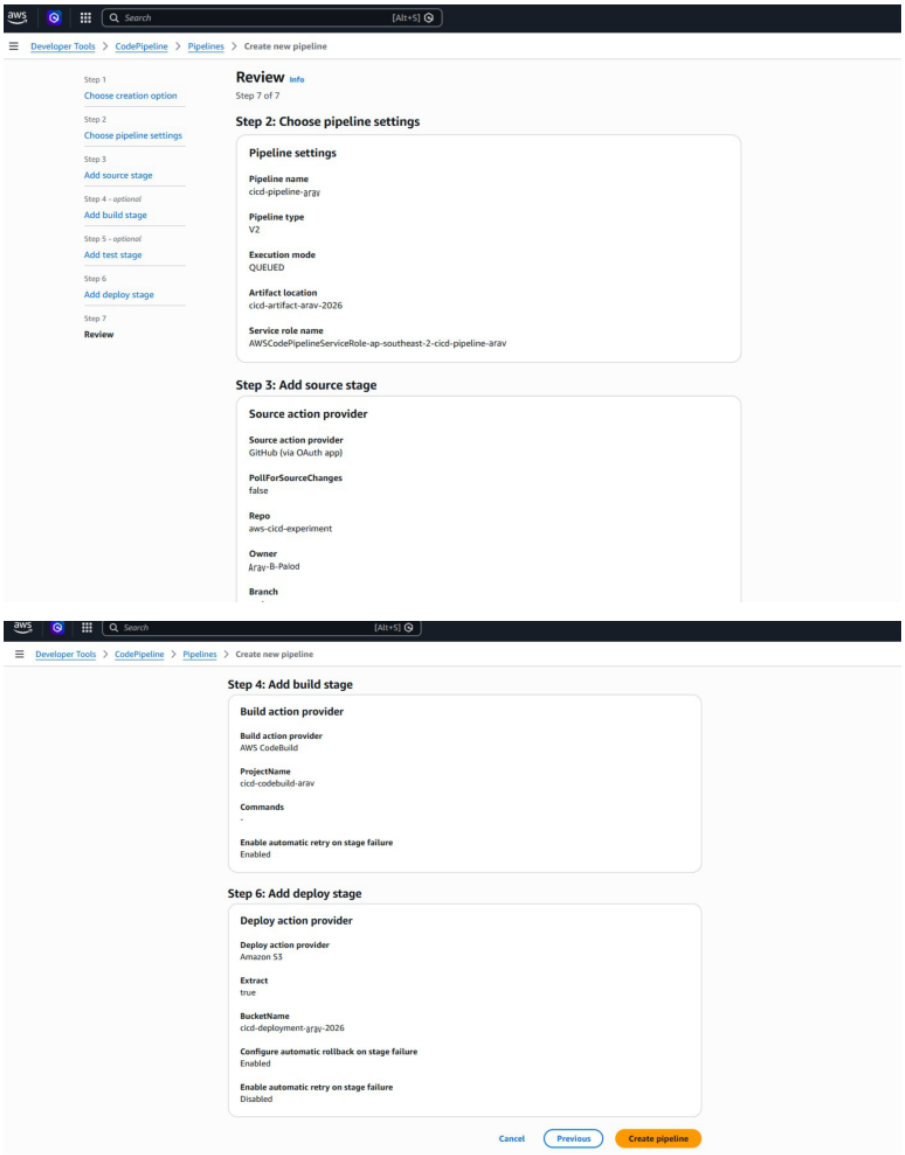


Experiment Evidence — Original Page 3

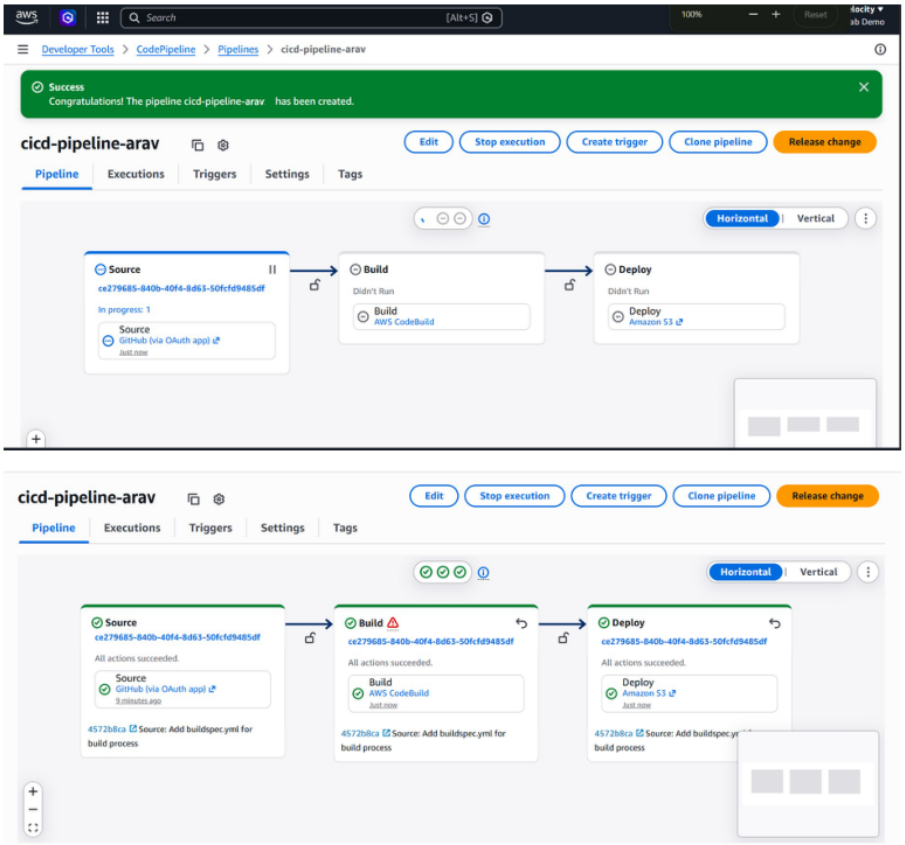




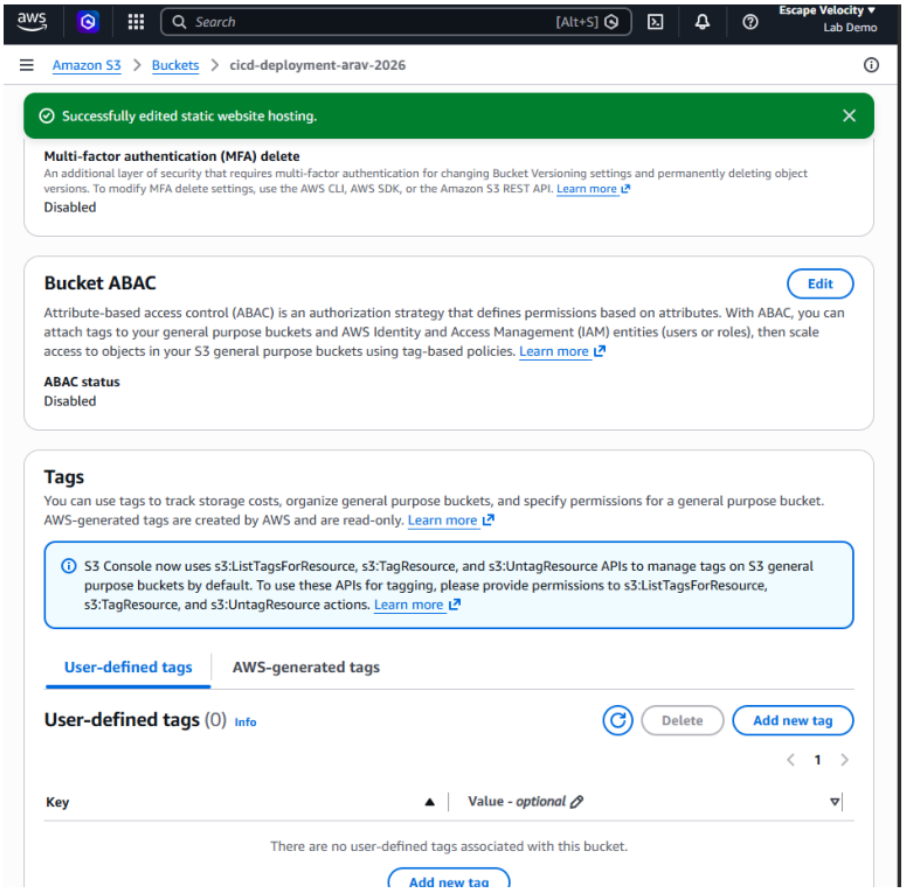
Experiment Evidence — Original Page 4



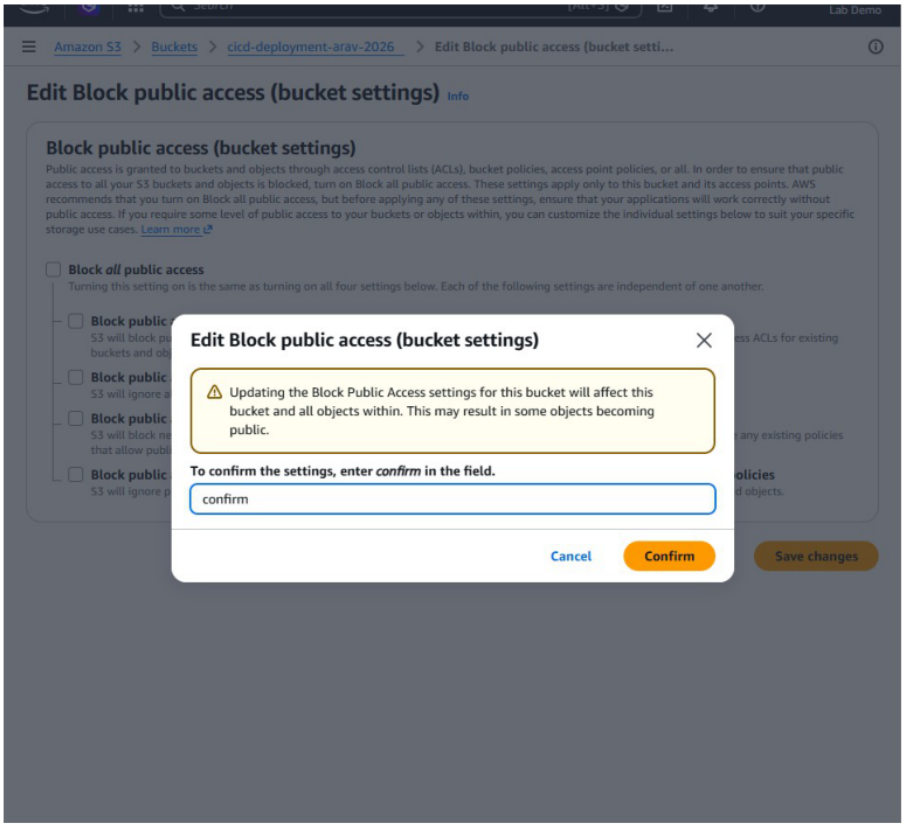
Experiment Evidence — Original Page 5

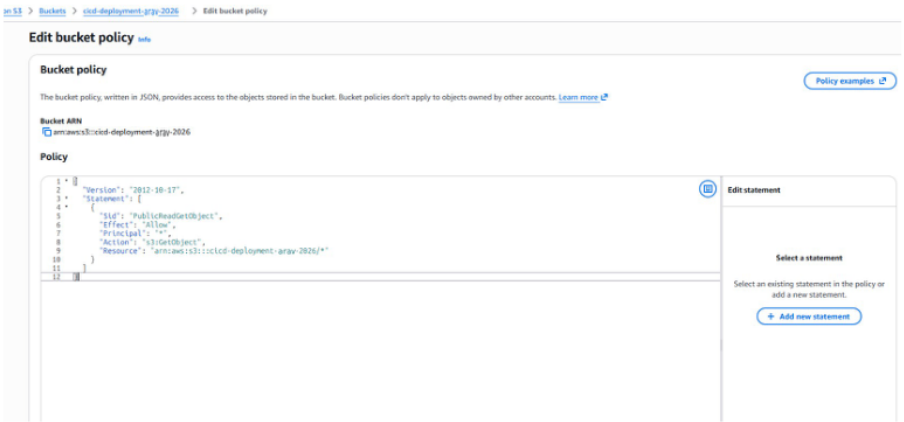


Experiment Evidence — Original Page 6



Experiment Evidence — Original Page 7

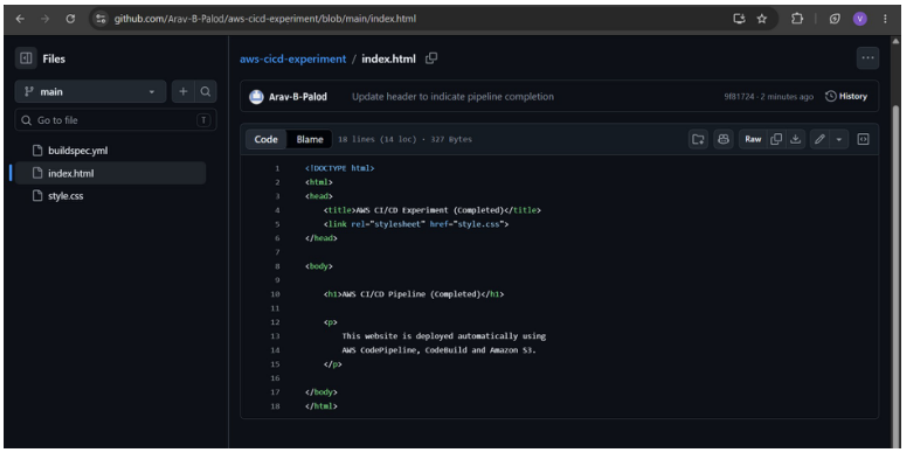


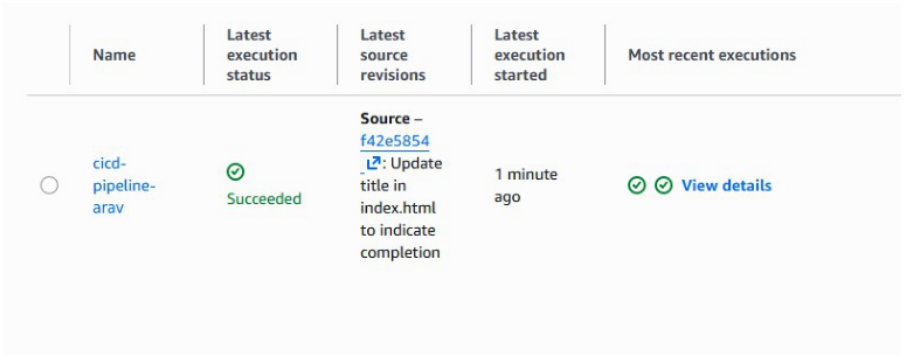


Experiment Evidence — Original Page 8



After Update:





Experiment Evidence — Original Page 9

